

CPT202 Software Engineering Group Project

Assignment 2 – Group Report and Presentation

Start date	28 th March 2026
Submission date	10 th May 2026
Assignment type	Group assignment
Percentage in final marks	Group Report 20% + Presentation 10% = 30%
Late submission policy	5% of the awarded marks shall be deducted for each working day after the submission date, up to a maximum of five working days. Late submission for more than five working days will not be accepted.
Important notes	• Plagiarism results in award of ZERO mark; all submissions will be checked by Turnitin. • The formal procedure for submitting coursework at XJTLU is strictly followed. Submission link on Learning Mall will be provided in due course. The submission timestamp on Learning Mall will be used to check late submission. • Generative AI Policy: The use of Generative AI for content generation is not permitted.

Learning Outcome

- A. Work effectively as part of a development team, demonstrating communication and interpersonal skills to collaboratively design and develop a software system.
- B. Explain the software development process, including methods and challenges in deploying systems to meet business goals.
- C. Specify and analyze the requirements of a software system using appropriate techniques.
- D. Produce and evaluate documentation that supports the software development lifecycle.
- E. Analyze legal, social, ethical, and professional issues involved in software development and deployment.

General instruction

By week 10, your team should have implemented 3 software increments. Therefore, this assignment requires your team to prepare and present the **first release** of the software system. At this stage, the team is expected to have completed the major system architecture, system-level software design, a set of core features, and relevant testing across the three increments.

In this assignment, your team is required to write a group report and a recorded group presentation.

- **Group Report:** The report must follow the format specified in the Group Report Template provided in the Assignment Package. Please note that the report must be no more than 15 pages, excluding appendices. Appendices may

include supporting materials such as diagrams, detailed testing records, deployment logs, and process evidence. All appendix materials must be clearly referenced and interpreted in the main report. Appendices alone do not constitute analysis.

- **Presentation:** The presentation should summarize the work undertaken to deliver the first software release and must be submitted as a 10-minute recorded video. The software demonstrated in the video must be accessed via the URL pre-allocated to your team. This URL must be clearly shown at the beginning of the demonstration. For verification purposes, the software at the submitted URL must remain unchanged for one week after submission.

The marking criteria and the weighting is based on the following factors:

A. Group Report

1. Report quality (15%):

The report should be well presented, with clear and concise language throughout. Where appropriate, images or diagrams should be used to support understanding. All visuals (e.g. architecture diagrams, screenshots, testing evidence) must be clearly labelled and explicitly referenced in the report text.

2. Introduction (15%):

The introduction must demonstrate a sufficient understanding of the project background and initial requirements.

The project background should clearly describe: the problem statement, project aims, user characteristics, project scope, assumptions and dependencies, and project risks.

The background should be presented from general to specific.

A case diagram should be included to help define and communicate the project scope. It must remain consistent with the stated requirements so as to avoid obvious omissions.

The introduction should also briefly explain how the initial requirements were translated into the first-release scope (e.g., prioritized features / PBIs), including the rationale used for prioritization.

3. Architectural Design (25%):

The architectural design should be clearly described, and the choice of architecture should be justified.

An overall architectural diagram must be presented to visually represent the system structure.

An ERD (Entity Relationship Diagram) of the database structure must also be provided, together with sufficient explanation of the design.

In addition to presenting the final architecture and ERD, briefly explain how the architecture supported the first release (e.g., modular

development, integration, and deployment), including any major architectural trade-offs.

4. Software Design and Engineering Practice (25%):

The report should show the high-level process flow of essential functions and explain how the software was modularised to support team development and integration.

Describe the supporting services and engineering tools actually used in this project (e.g., Git, Docker, Maven, CI tools, issue tracking), and explain with concrete examples how they supported day-to-day team collaboration and delivery, including task coordination, version control, building, testing, integration, and deployment.

The report should also explain the coding standards/conventions adopted and how the team applied or enforced them in practice.

In addition, it should describe the production environment and the team's deployment approach for the first release.

Analyze the legal, social, ethical, and professional issues involved in the development and deployment of the software and discuss how these issues influenced (or should influence) design decisions, implementation choices, testing, documentation, and deployment practices.

5. Software Testing (20%):

The report should describe how software testing was conducted, including: unit testing, integration testing, and acceptance testing.

The testing process, testing data, and results (including screenshots) should be included. Sufficient test data and test results must be provided.

In addition to reporting test types and results, the report should explain:

- when testing was conducted during development (e.g., before integration, after feature completion, before release/demo),
- how defects were tracked and resolved, and
- how testing outcomes informed bug fixing and/or design improvements before the first release.

B. Presentation

1. Understanding of the project (40%):

The presentation should demonstrate a good understanding of the project, including project background, requirements, architectural design, software design, database design, and the development process used to deliver the first release.

2. Software Demonstration (40%):

The software demonstration should cover all major aspects of the system, including front-end processes, back-end processes, and database access.

The demonstration should be presented through a realistic use case with close-to-real data and should emphasise the core workflow(s).

It should clearly show front-end/back-end interactions and database access. Where appropriate, key validation, error handling, or boundary cases should also be demonstrated.

3. Presentation Skills (20%):

The presentation should demonstrate effective communication at an appropriate pace. Any slides used should have a clear layout and appropriate use of visual effects. Figures should be presented clearly, and the presentation should be well timed.

Individual contribution

We recognize that individual team members may contribute differently to the group report and the presentation. Therefore, each team member is required to complete, sign, and submit an Individual Contribution Form, which is provided in the Assignment Package.

The contribution (%) stated in the form indicates the overall contribution of everyone to this assignment. The team leader should arrange a meeting to discuss and complete the form in the presence of all team members. The contribution percentage for each member must be agreed upon by all other members. If the team cannot reach agreement on contribution percentages, the team may seek assistance from the project supervisor.

Please note that only work completed by the assignment due date can be considered as contribution to this assignment. Routine discussion alone is not considered a contribution. However, documented process contributions (e.g., issue/task coordination, code review, test design, integration support, deployment setup) may be counted, if they are evidenced and agreed by the team.

Examples of contributions that may be counted include:

- Writing the report or presentation slides
- Presenting or demonstrating in the presentation
- Editing the report or video

The following are **not** considered contributions:

- Participating in discussions only
- Providing ideas and comments only
- Providing financial support

Submission and markings notes:

- Your team is expected to submit the following documents:
 - Group report (PDF format; 15 pages max)
 - Presentation slide (PDF format)
 - Presentation video (10 minutes max, 100MB max)
 - Source code in a **single** Zip file
 - Individual contribution form (PDF format)
- Name your documents accordingly:
 - Group report: CPT202-A2-Report-**GroupXXX**.pdf

- Presentation slide: CPT202-A2-Slide-**GroupXXX**.pdf
- Presentation video: CPT202-A2-Video-**GroupXXX**.mp4
- Source code: CPT202-A2-Code-**GroupXXX**.zip
- Individual contribution form: CPT202-A2-Contribution-**GroupXXX**.pdf
- Each team must submit ONLY one copy of the above documents by the team leader via Learning Mall
- When preparing your source code, take note that:
 - Files in proprietary or non-standard file formats may not be readable and so are likely to be ignored.
 - Do not submit the compiled binary, which will cause your zip file to be overly large.